

Microservice Architecture – Task 4

Link to GitHub

https://github.com/Romko-studik/micro_basics/tree/micro_mg

Architecture

The system includes three services:

- **facade-service:** Handles POST/GET HTTP requests from the client
- **logging-service:** Stores messages with deduplication and retry support
- **messages-service:** Returns messages

Running the Services

Run in separate terminals:

```
./start_services.sh
```

Tests:

```
./test.ch > result.log 2>&1
```

First of all from result.log we can see that all of messages were sent succesfully and after sending /get 3 times we can see that in total we get all of the messages sent:

```
Відправляємо 10 повідомлень через facade-service...
Відправляємо: msg1
Відправляємо: msg2
Відправляємо: msg3
Відправляємо: msg4
Відправляємо: msg5
Відправляємо: msg6
Відправляємо: msg7
Відправляємо: msg8
Відправляємо: msg9
Відправляємо: msg10

Повідомлення відправлені.

Детальніше разів отримуюмо 10'єднані повідомлення з facade-service...
Запит GET /get, спроба 1:
{"response":{"messages":{"msg1","msg5","msg9","msg4","msg2","msg6","msg3","msg7","msg8","msg10"}}}\n---\n{"messages":{"msg2","msg4","msg6","msg8","msg10"}}}\n---
...

Запит GET /get, спроба 2:
{"response":{"messages":{"msg1","msg5","msg9","msg4","msg2","msg6","msg3","msg7","msg8","msg10"}}}\n---\n{"messages":{"msg2","msg4","msg6","msg8","msg10"}}}\n---
...

Запит GET /get, спроба 3:
{"response":{"messages":{"msg1","msg5","msg9","msg4","msg2","msg6","msg3","msg7","msg8","msg10"}}}\n---\n{"messages":{"msg1","msg3","msg5","msg7","msg9"}}}\n---
...
```

Now lets look at logs of individual services, here are logging services:

```

    return await dependant.call(* values)
RuntimeWarning: Enable tracemalloc to get the object allocation traceback
Service 8071 - Received message: f8458082-ad7c-4b19-a9a8-c8e5f177383b -> msg6
INFO:      127.0.0.1:36976 - "POST /log HTTP/1.1" 200 OK
INFO:__main__:Service 8071 - Duplicate ignored: 617683c7-3567-4332-996d-15d6548d0a1a
Service 8071 - Received message: 617683c7-3567-4332-996d-15d6548d0a1a -> msg8
INFO:      127.0.0.1:36988 - "POST /log HTTP/1.1" 200 OK
INFO:__main__:Service 8071 - Duplicate ignored: 661585ef-17bd-4ac4-aa90-3a5c75044b5a
Service 8071 - Received message: 661585ef-17bd-4ac4-aa90-3a5c75044b5a -> msg9
INFO:      127.0.0.1:37004 - "POST /log HTTP/1.1" 200 OK

```

```

INFO:hazelcast_config:Stored message: 001c9923-ef62-4de3-8937-724071a42386 = msg1
INFO:__main__:Service 8072 - Stored: 001c9923-ef62-4de3-8937-724071a42386 -> msg1
INFO:__main__:Service 8072 - Total messages in map: 1
Service 8072 - Received message: 001c9923-ef62-4de3-8937-724071a42386 -> msg1
INFO:      127.0.0.1:53260 - "POST /log HTTP/1.1" 200 OK
INFO:hazelcast_config:Stored message: 9b68d5ec-d649-4f90-9af7-0fcd0b1c2295 = msg2
INFO:__main__:Service 8072 - Stored: 9b68d5ec-d649-4f90-9af7-0fcd0b1c2295 -> msg2
INFO:__main__:Service 8072 - Total messages in map: 2
Service 8072 - Received message: 9b68d5ec-d649-4f90-9af7-0fcd0b1c2295 -> msg2
INFO:      127.0.0.1:53272 - "POST /log HTTP/1.1" 200 OK
INFO:hazelcast_config:Stored message: 3986553f-2f5e-418d-8a47-2c22123ea7e5 = msg5
INFO:__main__:Service 8072 - Stored: 3986553f-2f5e-418d-8a47-2c22123ea7e5 -> msg5
INFO:__main__:Service 8072 - Total messages in map: 5
Service 8072 - Received message: 3986553f-2f5e-418d-8a47-2c22123ea7e5 -> msg5
INFO:      127.0.0.1:53284 - "POST /log HTTP/1.1" 200 OK
INFO:hazelcast_config:Stored message: 3cef780a-0436-4f8f-b908-1e47e90b0efc = msg6
INFO:__main__:Service 8072 - Stored: 3cef780a-0436-4f8f-b908-1e47e90b0efc -> msg6
INFO:__main__:Service 8072 - Total messages in map: 6
Service 8072 - Received message: 3cef780a-0436-4f8f-b908-1e47e90b0efc -> msg6
INFO:      127.0.0.1:53288 - "POST /log HTTP/1.1" 200 OK
INFO:hazelcast_config:Stored message: 6a4d383f-6ced-41b7-85a2-aed2691e0c2 = msg7
INFO:__main__:Service 8072 - Stored: 6a4d383f-6ced-41b7-85a2-aed2691e0c2 -> msg7
INFO:__main__:Service 8072 - Total messages in map: 7
Service 8072 - Received message: 6a4d383f-6ced-41b7-85a2-aed2691e0c2 -> msg7
INFO:      127.0.0.1:53304 - "POST /log HTTP/1.1" 200 OK
INFO:hazelcast_config:Stored message: 8dbbab60-23bb-4f52-9b3d-3a0f32c2b174 = msg10
INFO:__main__:Service 8072 - Stored: 8dbbab60-23bb-4f52-9b3d-3a0f32c2b174 -> msg10
INFO:__main__:Service 8072 - Total messages in map: 10
Service 8072 - Received message: 8dbbab60-23bb-4f52-9b3d-3a0f32c2b174 -> msg10
INFO:      127.0.0.1:53312 - "POST /log HTTP/1.1" 200 OK
INFO:__main__:Service 8081 - Retrieved all messages: 10 items
INFO:      127.0.0.1:53328 - "GET /all HTTP/1.1" 200 OK
INFO:__main__:Service 8081 - Retrieved all messages: 10 items
INFO:      127.0.0.1:53330 - "GET /all HTTP/1.1" 200 OK

```

```

INFO:hazelcast_config:Stored message: 73f62856-af72-435c-b006-b667deab0780 = msg3
INFO:__main__:Service 8073 - Stored: 73f62856-af72-435c-b006-b667deab0780 -> msg3
INFO:__main__:Service 8073 - Total messages in map: 3
Service 8073 - Received message: 73f62856-af72-435c-b006-b667deab0780 -> msg3
INFO: 127.0.0.1:52804 - "POST /log HTTP/1.1" 200 OK
INFO:hazelcast_config:Stored message: c3d86329-2de6-44cd-bf26-8a4dbd724481 = msg4
INFO:__main__:Service 8073 - Stored: c3d86329-2de6-44cd-bf26-8a4dbd724481 -> msg4
INFO:__main__:Service 8073 - Total messages in map: 4
Service 8073 - Received message: c3d86329-2de6-44cd-bf26-8a4dbd724481 -> msg4
INFO: 127.0.0.1:52806 - "POST /log HTTP/1.1" 200 OK
INFO:hazelcast_config:Stored message: 9fbf2cd1-e3ff-4584-b4e1-78b507847da3 = msg8
INFO:__main__:Service 8073 - Stored: 9fbf2cd1-e3ff-4584-b4e1-78b507847da3 -> msg8
INFO:__main__:Service 8073 - Total messages in map: 8
Service 8073 - Received message: 9fbf2cd1-e3ff-4584-b4e1-78b507847da3 -> msg8
INFO: 127.0.0.1:52820 - "POST /log HTTP/1.1" 200 OK
INFO:hazelcast_config:Stored message: ca2533b8-f19f-4186-a812-a3119e01adb4 = msg9
INFO:__main__:Service 8073 - Stored: ca2533b8-f19f-4186-a812-a3119e01adb4 -> msg9
INFO:__main__:Service 8073 - Total messages in map: 9
Service 8073 - Received message: ca2533b8-f19f-4186-a812-a3119e01adb4 -> msg9
INFO: 127.0.0.1:52828 - "POST /log HTTP/1.1" 200 OK
INFO:__main__:Service 8081 - Retrieved all messages: 10 items
INFO: 127.0.0.1:52830 - "GET /all HTTP/1.1" 200 OK

```

message services:

```

INFO: Uvicorn running on http://0.0.0.0:8010 (Press CTRL+C to quit)
[messages-service] Stored: {'id': '001c9923-ef62-4de3-8937-724071a42386', 'msg': 'msg1'}
[messages-service] Stored: {'id': '73f62856-af72-435c-b006-b667deab0780', 'msg': 'msg3'}
[messages-service] Stored: {'id': '3986553f-2f5e-418d-8a47-2c22123ea7e5', 'msg': 'msg5'}
[messages-service] Stored: {'id': '6a4d383f-6ced-41b7-85a2-aed2691e0c2', 'msg': 'msg7'}
[messages-service] Stored: {'id': 'ca2533b8-f19f-4186-a812-a3119e01adb4', 'msg': 'msg9'}
INFO: 127.0.0.1:43424 - "GET /message HTTP/1.1" 200 OK

```

```

INFO: Uvicorn running on http://0.0.0.0:8011 (Press CTRL+C to quit)
[messages-service] Stored: {'id': '9b68d5ec-d649-4f90-9af7-0fcd0b1c2295', 'msg': 'msg2'}
[messages-service] Stored: {'id': 'c3d86329-2de6-44cd-bf26-8a4dbd724481', 'msg': 'msg4'}
[messages-service] Stored: {'id': '3cef780a-0436-4f8f-b908-1e47e90b0efc', 'msg': 'msg6'}
[messages-service] Stored: {'id': '9fbf2cd1-e3ff-4584-b4e1-78b507847da3', 'msg': 'msg8'}
[messages-service] Stored: {'id': '8dbbab60-23bb-4f52-9b3d-3a0f32c2b174', 'msg': 'msg10'}
INFO: 127.0.0.1:41182 - "GET /message HTTP/1.1" 200 OK
INFO: 127.0.0.1:41190 - "GET /message HTTP/1.1" 200 OK

```